Barclays Transactional Data Analytics Project

Project Overview

Analyze transactional data from Barclays to uncover customer behavior, channel preferences, fraud risk, revenue trends, and operational patterns. The goal is to provide actionable insights and recommendations that can inform business decisions and improve customer experience.

Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load, Inspect & Cleaning Data

Load the transactional data into a DataFrame and perform initial inspection to check dimensions, data types, and missing values. Clean the data by dropping rows with critical missing values, filling missing values with appropriate defaults, and ensuring data types are correct for analysis.

```
df = pd.read_csv('/content/drive/MyDrive/Barclays Transactional Data.csv')
df.head()
```

```
   transaction_id  customer_id  customer_age  transaction_date  ...  \
0               1         1001            45        2024-01-01  ...
1               2         1002            32        2024-01-02  ...
2               3         1003            28        2024-01-03  ...
3               4         1004            51        2024-01-04  ...
4               5         1005            39        2024-01-05  ...

   channel  customer_publication_date  customer_publication_time  \
0  Online                 2024-01-01                   0.000000
1  Online                 2024-01-02                   0.000000
2  Online                 2024-01-03                   0.000000
3  Online                 2024-01-04                   0.000000
4  Online                 2024-01-05                   0.000000

   customer_publication_duration  ...  \
0                              0  ...
1                              0  ...
2                              0  ...
3                              0  ...
4                              0  ...

   customer_publication_duration_bucket  ...  \
0                                    0  ...
1                                    0  ...
2                                    0  ...
3                                    0  ...
4                                    0  ...

   customer_publication_duration_bucket_encoded  ...  \
0                                             0  ...
1                                             0  ...
2                                             0  ...
3                                             0  ...
4                                             0  ...

   customer_publication_duration_bucket_encoded.1  ...  \
0                                               0  ...
1                                               0  ...
2                                               0  ...
3                                               0  ...
4                                               0  ...

   customer_publication_duration_bucket_encoded.2  ...  \
0                                               0  ...
1                                               0  ...
2                                               0  ...
3                                               0  ...
4                                               0  ...

   customer_publication_duration_bucket_encoded.3  ...  \
0                                               0  ...
1                                               0  ...
2                                               0  ...
3                                               0  ...
4                                               0  ...
```